

ROUND 1 · VIRTUAL PITCH · TEAM BU 1

ARBITER Master Playbook

The demo, the validation, the business case, and the regulatory story, in one document.

Everything decided across the concept phase, the mentor call, and the official Round 1 guidelines, resolved into a concrete build plan. Written to be executed, not admired. The demo specification in Part 3 is the priority: it is the deliverable that Round 1 is actually judged on.

Submission due	Presentation	Slots	Team
August 16, 11:59 PM ET	15 min + 5 min Q&A	August 17 to 19	Jack He, Andres Lopez, Jose Cruz-Lopez

PART 0 What Just Changed

01 Corrections From the Official Guidelines

The Round 1 guidelines materially change the plan. Six things are different from what we were building toward, and two of them are large.

Item	What we assumed	What is actually required
Format	A 5-minute video pitch	15-minute live presentation plus 5-minute Q&A with a live judging panel. Three times the runway, and a real Q&A to survive. Our 5-minute script is now the spine of a much larger talk, not the talk itself.
Demo	Possibly expected	Required. Submission must include a functioning demo, and the guidelines name Lovable and Claude as acceptable mediums. This settles the build question completely.
Round name	Round 2	Officially Round 1 . The video was the Preliminary Round. Use the official name in all materials.
Judging criteria	Innovation, Pfizer Fit, Technical Feasibility, Non-Technical Feasibility, Presentation	Now: Innovation, Pfizer & Patient Impact , Technical Feasibility, Business & Operational Feasibility , Presentation Quality. Patient impact is new and our materials do not address it at all. Business feasibility is now its own named criterion.
Deadline	Week of August 17	Materials due August 16 at 11:59 PM ET to the ambassador. Presentations August 17 to 19. Finalists announced August 26.
Final round	Virtual	In person at Pfizer HQ in New York City. At least one team member must attend. Worth planning for now.

The two that change our work most

Patient impact is now a judged criterion and we have never written a word about it. Part 8 fixes this. It is a straightforward story for us and it is one of the strongest angles we have, because our own case study involves three trial participants who met Hy's Law criteria for serious liver injury.

Fifteen minutes is a completely different presentation. The recommended structure has eight sections. Our five-minute script covers roughly the first half. Part 9 lays out the full 15-minute structure mapped to the guidelines' recommended order.

What the guidelines say judges want to leave understanding

Verbatim from the FAQ: how the solution works, what technologies or processes may be involved, key assumptions, and potential challenges. And explicitly: teams are not expected to provide production-ready technical designs. That is permission to keep the deck at the level of clear explanation rather than engineering detail, which aligns exactly with what our mentor advised.

The framing sentence the guidelines ask for

The guidelines say the strongest teams articulate: "We help [user] do [action] so they can achieve [outcome]." Ours is: **We help preclinical safety leads reason through conflicting toxicity evidence so they can make consistent, defensible go or no-go decisions.** Put this on an early slide, in these words.

02 What Our Mentor Told Us

The call produced clear direction. These are the instructions, extracted and ranked by how much they change the work.

Direction	What we do about it
Focus on the business side	Asked directly whether to go business or technical, the answer was business. Knowing how it works, who uses it, who works with it, and what it makes better in the business. The deck goes business-heavy; the technical depth moves into the demo.
Be less technical in the slides	If we go technical, go lighter. Focus on one or two cases. The demo carries the technical proof, so the slides do not have to.
The MVP is still too broad	Direct feedback: we narrowed it, but are still covering too much too quickly. The instinct given was one toxicity area and a small number of competing evidence sources, rather than liver plus cardiac plus regulatory reports plus full AI governance. Part 3 scopes to exactly that.
Prove it beats the simple average	Explicitly endorsed: it is not only that it produces a recommendation, but how you prove that recommendation is better than a simple average approach. He said this would be especially good to show in the presentation. This is our headline demo moment.
ARBITER does not decide alone	Called out as central to the story. Decision support and evidence reconciliation, never replacement.
Map the regulatory landscape	He asked us to create a landscape of the laws we would operate under, and to think about what the regulator would say and do. Part 7 delivers this.
Simple and concise	Repeated several times. Focus on what we are offering, not general context.

The one-liner he gave us, which is better than ours

Asked what a judge should remember if they remember only one thing, his answer was: **ARBITER helps toxicologists reason through conflicting safety signals, transparently, with the human experts still making the decision.** That single sentence contains the user, the problem, the differentiator, and the safety guarantee. Use it as the closing line and the deck subtitle.

The correction hidden in his regulatory comment

He noted that you never know what a regulator will want, so what you actually need is something ready for **scientist review**, and after the scientist reviews it, then it goes to the regulator. This is a real positioning correction. We have been saying regulator-ready dossier. We should say **review-ready evidence package**: the artifact a scientist signs off on, which then feeds the regulatory submission. It is more accurate, more modest, and more credible to anyone who has worked near a submission.

Logistics fixed on the call

- Presentation slot requested: Wednesday, 11:00 AM to 2:00 PM, with 2:00 to 4:00 PM as backup. Form submitted.
- Roughly 22 teams in the round. His guess at advancement was around half, but he was explicit that he did not know. Treat as unknown.
- Next check-in: same time next week, with the demo and the business presentation to show.

PART 1 The Idea, Locked

03 Positioning and Message

The locked messaging stack

One line, for the judge who remembers one thing: ARBITER helps toxicologists reason through conflicting safety signals, transparently, with the human experts still making the decision.

The guidelines format: We help preclinical safety leads reason through conflicting toxicity evidence so they can make consistent, defensible go or no-go decisions.

The differentiator: Everyone else is building tools to predict toxicity. ARBITER reasons through the conflicts between those predictions. It is not another predictor. It is the layer that decides.

Three rules for every asset

- **Lead with consistent and defensible, never with fast.** Both our mentor and our own stress-testing landed here independently. Speed is a side effect, not the pitch.
- **Never imply replacement.** The human expert makes the decision. Say it in the opening and again in the close.
- **Say review-ready evidence package**, not regulator-ready dossier.

What ARBITER is, in five moves

Gather the conflicting evidence. Reconcile it with explicit logic that explains why one piece outranks another. Quantify the remaining uncertainty as an honest range instead of a false-precision number. Ground the verdict in the known biological mechanism and name the one experiment that would resolve the doubt. Document the reasoning so a scientist who was not in the room can reconstruct the decision.

04 The Case That Carries the Whole Pitch

TAK-994 is the most valuable asset in this project. It is a real, published, verifiable failure that our system

is designed to catch, and it now doubles as our patient-impact story.

What happened, verified against the primary literature

TAK-994 was an oral orexin receptor 2 agonist developed by Takeda for narcolepsy type 1, and the first of its class to reach clinical trials. Both rat and non-human primate repeat-dose toxicity studies failed to detect any hepatotoxicity liability before first-in-human dosing. Phase 1 in healthy volunteers showed no signs of liver injury. In the Phase 2 trial, 73 patients with narcolepsy type 1 received placebo or TAK-994 over eight weeks. Efficacy was strong and statistically significant across all doses. But eight patients exceeded predefined liver enzyme thresholds, and **three met Hy's Law criteria**, the clinical marker for serious drug-induced liver injury. The trial was stopped and the program was terminated.

Murine studies initiated *during* the clinical trial, too late to prevent exposure, revealed hepatic single-cell necrosis following cytochrome P450 induction at clinically relevant doses. Meanwhile the standard in vitro panels showed wide safety margins, more than 100-fold, against the known intrinsic DILI mechanisms including cytotoxicity, mitochondrial toxicity, and bile salt export pump inhibition.

Why this is the perfect ARBITER case

The evidence genuinely conflicted across sources, which is exactly the situation ARBITER is built for. Rat said clean. Primate said clean. In vitro panels said wide margins. Mouse said toxic, but only once someone thought to run it, during the trial. The species that carried the signal was not among the two required species.

And note what the standard paradigm asked for: ICH M3 and S9 require repeat-dose toxicity studies in two mammalian species, one non-rodent, before first-in-human dosing. TAK-994 satisfied that requirement completely and still injured patients. The failure was not non-compliance. It was that the evidence integration step, deciding what a set of clean animal studies and wide in vitro margins actually license you to conclude, has no structured method behind it.

HOW TO SAY IT IN THE ROOM

TAK-994 passed every study the regulations required. Rat clean, primate clean, in vitro margins over a hundredfold. Three patients still ended up with serious liver injury. The mouse study that found the mechanism was run during the trial, after the harm. Nobody broke a rule. The rules just do not tell you what to do when the evidence disagrees.

PART 2 The Demo (Build This First)

05 Demo Scope and Architecture

A functioning demo is now a required submission item. This is the deliverable Round 1 turns on, and it is where our technical credibility is either earned or lost. Build this before anything else.

The scope, narrowed as instructed

One toxicity area: hepatotoxicity. Four evidence streams. A small set of compounds where the evidence genuinely conflicts. Two pipelines compared side by side. Nothing else.

Element	Decision
Endpoint	Drug-induced liver injury only. No cardiac, no mutagenicity, no multi-organ.
Ground truth	FDA DILrank classifications. This is the answer key.
Evidence streams	Four, fixed: structural alert or QSAR prediction, hepatocyte cytotoxicity assay, toxicogenomic signature, mechanistic transporter assay such as bile salt export pump inhibition.
Compounds	A curated set where the four streams disagree. Enough to show a pattern, few enough to explain. Report the exact number honestly.
The comparison	Pipeline A: naive aggregation and a plain language model given identical evidence. Pipeline B: ARBITER with argumentation and evidence fusion.
Explicitly out	Other organs, proprietary data, live lab integration, full dossier generation, multi-user accounts, real-time data pulls.

Build medium

The guidelines name Lovable and Claude as acceptable. Given the three-and-a-half week window and that judges need to click through it live, build a self-contained web application with the reasoning results precomputed and cached. **Do not make a live API call the critical path of a judged demo.** Precompute every compound's full result, ship the data with the app, and let the interface reveal it step by step. The reasoning engine still runs for real when you generate the results; the demo just does not depend on a network call at the moment a judge is watching.

Demo failure modes to engineer against

Live model calls that time out. Screen-share lag on animated transitions. A dataset too large to load. Anything requiring a login. Any dependency on our own laptop environment. The demo should open in a browser from a link, work offline, and run identically for anyone who clicks it. Build a recorded walkthrough as a fallback and submit it alongside, since the guidelines explicitly allow a recorded walkthrough.

06 The Demo, Screen by Screen

Roughly four minutes of the 15-minute presentation. Every screen has one job. Nothing on screen that does not serve the argument.

1 Compound selection

15 SEC

A short list of compounds, each tagged with whether the evidence agrees or conflicts. The judge picks one, or we drive to a chosen case. Conflict cases are visually marked.

The point: establishes immediately that conflict is common, not exotic. Show the count: of the compounds we assembled, this many had at least one source disagreeing with another.

2 The evidence panel

30 SEC

Four cards, one per stream, each showing its verdict, its confidence, and its provenance. The conflict is visible at a glance through color: some say toxic, some say clean, one is ambiguous.

The point: this is what lands on a safety lead's desk. Four credible sources, disagreeing, with no built-in method for resolving them.

3 Pipeline A: what happens today

30 SEC

Run the naive approach. Majority vote or simple averaging returns a verdict, and a plain language model given identical evidence returns its own verdict with a confidence number. Show both.

The point: the aggregate answer is confident, unexplained, and on our chosen case, wrong. The language model produces a number nobody can interrogate.

4 Pipeline B: ARBITER's argument trace

70 SEC

The centerpiece. ARBITER's reasoning renders as a readable chain: this evidence was admitted, this one was defeated, by this named rule, therefore this verdict. Each step cites the rule that produced it and the framework the rule came from.

The point: this is the product. Linger here. A judge should be able to read the reasoning and agree or disagree with it, which is impossible with any confidence score. Say out loud: you are not being asked to trust this, you are being asked to read it.

5 The honest uncertainty range

25 SEC

A belief-to-plausibility bar rather than a single percentage. The width of the gap is the system's ignorance, made visible.

The point: we never emit a fake ninety-five percent. When evidence is thin, the bar widens, and that is the system telling you it does not know enough.

6 Counterfactual: what would flip this

30 SEC

For the current verdict, highlight which single piece of evidence, if reversed, would change the outcome. One click, one highlighted card.

The point: explainability and experiment planning in one view. This is the feature no competing tool has, and it makes the reasoning tangible in a way a static trace does not.

7 The recommended next experiment

20 SEC

Given the unresolved uncertainty, ARBITER names the single assay that would most reduce it, with the reason.

The point: it does not just judge, it tells you how to resolve the doubt. This is the rescue path for borderline candidates, and it is where cost savings come from.

8 Ground truth reveal

20 SEC

Reveal the DILrank classification. On the chosen case: naive aggregation was wrong, ARBITER was right, and you can see exactly why.

The point: the dramatic beat. Do not oversell it. One case is an illustration, not proof. The proof is the next screen.

9 The consistency test

40 SEC

Run the same compound through both pipelines several times. The language model's verdict or confidence varies between runs. ARBITER returns an identical verdict every time, because symbolic reasoning is deterministic.

The point: this is our strongest claim and it maps directly onto the word our mentor and ambassador both used, consistent. Same evidence, same answer, every time, with the same explanation. That is what makes a decision defensible.

THE HONESTY LINE, SAY IT DURING THE DEMO

Everything you are seeing runs on public data, and our results are computed on a benchmark set, not asserted. One case is an illustration. The aggregate numbers on the next slide are the actual claim.

07 The Reasoning Engine: What to Actually Implement

Concrete enough to hand to Jack and Andres and start coding. Two components: the preference ordering that resolves conflicts, and the fusion that quantifies what remains uncertain.

Component 1: the preference ordering

These rules decide which evidence wins. They are not our opinions. Each traces to an established framework, and each must be written down and timestamped before we look at any outcomes, so nobody can accuse us of fitting them to the answer key.

R1 · HUMAN RELEVANCE

Human-cell evidence defeats animal in vivo evidence when the question is human hepatotoxicity.

Grounded in the FDA's own NAMs direction and in the TAK-994 case, where rat and primate were both clean and humans were injured.

R2 · MECHANISTIC PROXIMITY

Evidence that directly measures a key event in the adverse outcome pathway defeats evidence that only correlates with structure.

Follows from the OECD Adverse Outcome Pathway framework, where key event relationships carry explicit confidence levels.

R3 · EXPOSURE RELEVANCE

A positive finding at clinically relevant exposure defeats a negative finding whose exposure margin is unstated or untested at that range.

Directly from TAK-994: the murine necrosis appeared at clinically relevant doses while in vitro margins were reported as over a hundredfold at different conditions.

R4 · APPLICABILITY DOMAIN

Evidence from a model operating outside its applicability domain is admitted with reduced weight, or excluded.

Standard practice in QSAR regulatory acceptance. A prediction about a compound unlike the training set is not the same kind of evidence.

R5 · STUDY RELIABILITY

Higher-reliability studies defeat lower-reliability ones at equal mechanistic relevance.

Klimisch scoring, the standard reliability system used in regulatory toxicology submissions.

R6 · CONCORDANCE

Independent sources agreeing raises confidence more than a single source agreeing with itself.

Basic weight-of-evidence logic, and it is what the fusion layer formalizes.

Every rule is editable in the interface. That is the point: a toxicologist can open the ordering, disagree, change it, and watch the verdict update. Demonstrating that live, even briefly, is worth more than describing it.

Component 2: uncertainty fusion

Each evidence stream assigns its confidence across three possibilities rather than two: mass toward toxic, mass toward safe, and mass left uncommitted, which represents what that source genuinely cannot tell you. Combining across sources yields two numbers per compound: **belief**, the evidence that positively supports the conclusion, and **plausibility**, everything not actively contradicted. The gap between them is the honest uncertainty.

A source that is silent on a compound contributes uncommitted mass rather than being treated as a vote for safe. That distinction is the entire reason a fusion approach beats averaging, and it is worth one sentence in the presentation: **absence of evidence is not evidence of absence, and averaging treats it as though it were.**

Implementation order for the three of us

1. **Data layer first.** Assemble compounds, match identifiers across sources by chemical structure, and normalize each stream to a common signal scale. This is the hidden time sink. Start here.
2. **Conflict detection.** Formally define disagreement in code and label the conflict subset.
3. **The rules engine.** Implement R1 through R6 as explicit, inspectable rules with a trace output. The trace is a first-class output, not a log.
4. **Fusion.** Belief and plausibility per compound.
5. **Baselines.** Majority vote, best single source, and the language model pipeline.
6. **Interface last.** The nine screens, over precomputed results.

PART 3 Validation and Metrics

08 Formal Validation Methodology

Our mentor said it plainly: it is not only that it produces a recommendation, but how you prove that recommendation is better than a simple average. This section is that proof, designed before we look at

any results.

The three baselines

Baseline	What it tests
Majority vote	The naive aggregation our mentor named. Does structured reasoning beat counting votes?
Best single source	The honest bar. If one predictor alone matches the whole system, our complexity is not earning its place. Include this even though it is the least flattering.
Language model, no symbolic layer	The ablation. Identical evidence, asked for a verdict and confidence. This isolates the contribution of the reasoning engine itself.

Why the ablation is the most important thing we build

Our entire thesis is that explicit symbolic reasoning outperforms black-box judgment on conflicting evidence. Without the ablation, that is an argument. With it, it is a measurement. It is also the single most impressive thing we can say in a judged Q&A: we designed a test that could have disproved our own premise, and we ran it.

Prepare for a mixed result. If the language model matches us on accuracy, our advantage is consistency, traceability, and calibrated uncertainty, which are likely the real advantages anyway. Report whatever we find. A team presenting an honest mixed result with a clear interpretation is more credible than one presenting a suspiciously clean sweep.

Protocol

- **Pre-register the rules.** Write R1 through R6 down, timestamp the file, commit it before any evaluation. Show the timestamp in the presentation if challenged. This single practice answers the sharpest possible attack, which is that we tuned the rules to fit DILrank.
- **Hold out a test set** that is untouched during development.
- **Report on the conflict subset separately.** Overall accuracy is inflated by easy cases where every source agrees. The conflict subset is where the product lives.
- **Repeat runs for the stochastic pipeline.** The language model baseline gets run multiple times per compound to measure variance. ARBITER gets the same treatment to demonstrate it does not vary.
- **Report confidence intervals and the sample size.** Our n is small. Say so. Understating our own certainty is on-brand for a project about honest uncertainty.

09 Success Metrics

Four metrics, in priority order, each tied to something a judge cares about.

Metric	Definition	Why it matters
Conflict-subset accuracy	Balanced accuracy against DILrank, computed only on compounds where sources disagree, versus all three baselines.	The headline. This is the product's actual job.

Metric	Definition	Why it matters
Decision consistency	Proportion of repeated runs returning an identical verdict, ARBITER versus the language model pipeline.	Maps directly to the word both our mentor and ambassador used. Likely our strongest result.
Uncertainty calibration	How often the true label falls within the belief-to-plausibility interval, reported alongside mean interval width.	Proves the honest-range claim is real rather than rhetorical. A wide-but-always-right interval is useless, so both numbers must be shown.
Abstention quality	Accuracy on cases where ARBITER issues a verdict, versus cases it declines, and the decline rate.	Demonstrates the safety behavior: it should be more accurate on what it commits to, and route the rest to a human.

Operational metrics for the business case

These are modeled rather than measured at this stage, and must be labeled as such: time to assemble a review-ready evidence package versus manual assembly, number of confirmatory assays avoided through targeted next-experiment recommendations, and inter-reviewer variance removed. Present as projected operational impact with the assumptions stated, never as measured results.

PART 4 The Business Case

10 Business Impact, Quantified

Business and Operational Feasibility is now its own judged criterion, and our mentor told us to lead with the business side. This is the arithmetic, with every source named and every assumption exposed.

The cost baseline

Figure	Source
Capitalized cost per approved drug averaged approximately \$2.23 billion in 2024 for major biopharma; the widely cited Tufts figure is \$2.6 billion in 2013 dollars	Tufts Center for the Study of Drug Development; 2024 industry analysis
Roughly 10 to 14 percent of drugs entering Phase 1 reach approval	Wong et al. and Tufts phase-transition data
Average Phase 3 trial cost estimated at \$255 million; median pivotal trial cost estimated at \$48 million with an interquartile range of \$20 to \$102 million	Tufts 2014; Institute for Safe Medication Practices analysis
Median per-phase costs of approximately \$3.4 million (Phase 1),	Martin et al. 2017

Figure	Source
\$8.6 million (Phase 2), \$21.4 million (Phase 3)	
Toxicity accounts for roughly one third of candidate attrition; DILI is a leading cause of both development termination and post-market withdrawal	Multiple reviews; DILI literature consensus

Be careful with these numbers in the room

The cost-per-drug figures are contested and vary by an order of magnitude depending on methodology, mostly over whether capital costs and failure costs are included. Present a range, name the source, and say the estimates vary. A Pfizer judge will know this literature far better than we do, and acknowledging the controversy is safer and more credible than quoting one number as fact.

The value model, stated as arithmetic a judge can check

ARBITER's value is **cost avoidance through earlier and better decisions**, in three channels.

CHANNEL 1: CATCH EARLIER

Moving a toxicity-driven termination from Phase 2 to preclinical avoids the clinical spend that follows. Using conservative median per-phase costs, that is millions per event; using capitalized industry figures, substantially more. The TAK-994 program consumed full preclinical, Phase 1, and a completed Phase 2 before termination.

CHANNEL 2: RESCUE BORDERLINE CANDIDATES

A conservative manual process kills ambiguous candidates out of caution. ARBITER instead names the one experiment that resolves the ambiguity. Every viable candidate rescued is an asset that would otherwise have been written off.

CHANNEL 3: FEWER WASTED ASSAYS

Targeted next-experiment recommendations replace running several confirmatory assays hoping one resolves the question. Direct, immediate operational savings.

THE COST SIDE

No foundation model training. No new lab infrastructure. Runs on data already generated and infrastructure already owned. Cost is engineering time plus expert time to curate and validate the rule base. Operational, not capital.

THE ASYMMETRY, WHICH IS THE WHOLE BUSINESS CASE

The build is measured in engineering months. A single avoided late-stage toxicity failure is measured in tens to hundreds of millions. We do not need to claim a large effect size for the arithmetic to work. We need to claim a real one.

What kind of business is this

Be direct about this, because a judge will ask. ARBITER is an **internal capability, not a product to sell**. Its value to Pfizer is avoided cost, faster and more consistent decisions, and a stronger regulatory position, not licensing revenue. That framing is honest and it fits the hackathon, where the intellectual property is Pfizer's. If pressed on longer-term commercial potential, the credible extensions are a cross-company consortium model

for shared safety reasoning standards, or contribution to the emerging NAMs qualification ecosystem. Offer those as possibilities, not as a plan.

11 Adoption, Cost, and Scalability

The guidelines name adoption, cost, scalability, compliance, and operational constraints explicitly. Have an answer for each.

Dimension	Our position
Adoption	Shadow mode first: runs alongside the existing review, output reviewed but never binding, proving its record against decisions already made. Then advisory, where its evidence package becomes the pre-read. Then integrated. The human decides at every stage.
Cost	No model training, no lab infrastructure, no new data generation. Engineering and expert curation time. Deployable on infrastructure Pfizer already runs.
Scalability	Adding an endpoint means adding evidence streams and pathway-specific rules, not rebuilding. The reasoning engine is endpoint-agnostic. Liver first, then cardiac, then others.
Compliance	Self-hosted on open-weight models so proprietary compound data never leaves the environment and never trains an external model. Full audit trail supports electronic-records expectations.
Operational constraints	Reads from existing lab notebook, information management, and assay systems. Nobody changes how data is captured. Nothing is ripped out.
Data availability	The MVP uses only public data, which the guidelines explicitly permit. Internal deployment uses data Pfizer already generates.

PART 5 Regulatory and Workflow

12 The Regulatory Landscape

Our mentor asked us to map the laws we would operate under and to think about what a regulator would say. This landscape is unusually favorable, and one item is very recent and very specific to us.

Instrument	Date	What it does, and why it matters to ARBITER
FDA Modernization Act 2.0	2022	Clarified in law that non-animal alternatives can support an investigational new drug application in lieu of animal studies. The legal foundation everything else rests on.
ICH M3(R2) and S9	Standing	Require repeat-dose toxicity studies in two mammalian species, one non-rodent, before first-in-human. This is the paradigm TAK-994 satisfied and still failed, which is exactly our argument.
FDA Roadmap to Reducing Animal	April 2025	Agency-wide strategy to phase down animal testing by promoting new approach methodologies, naming AI models, organ-on-chip, and cell-based assays explicitly.

Instrument	Date	What it does, and why it matters to ARBITER
Testing in Preclinical Safety Studies		
FDA draft guidance: General Considerations for the Use of NAMs in Drug Development	March 2026	The key one. Provides a validation framework and regulatory expectations for non-animal methods. Critically, it indicates that fit-for-purpose NAMs may be reviewed without full formal validation <i>when integrated into a weight-of-evidence assessment</i> . Structured weight-of-evidence integration is precisely what ARBITER produces.
CDER streamlined nonclinical studies and acceptable NAMs	Ongoing	An inventory of contexts where CDER is open to streamlined nonclinical programs, with review divisions assessing whether weight-of-evidence risk assessment is sufficient based on totality of evidence.
Monoclonal antibodies: streamlined nonclinical safety studies	2025	Recommends reducing long-term primate studies for certain antibodies using computational models and human-relevant systems with a weight-of-evidence assessment.
EMA NAM qualification and Innovation Task Force	Ongoing	Parallel European pathway, also centered on weight of evidence. Both FDA and EMA platforms indicate no regulatory barrier to submitting NAM data given adequate qualification.
EU animal testing phase-out roadmap	Expected 2026	A legislative shift from reduction toward replacement where scientifically justified.
OECD IATA and Adverse Outcome Pathway framework	Standing	The international framework for integrated testing and assessment, and the source of the mechanistic confidence levels our rules rely on.

The regulatory argument, in one paragraph

Regulators are actively moving away from animal studies as the default source of safety evidence, and toward human-relevant methods integrated through weight of evidence. That shift creates a problem nobody has solved: when you replace two animal species with a portfolio of in vitro, in silico, and omics methods, you get more evidence streams and therefore more conflict, with no structured way to reconcile them. The March 2026 draft guidance makes weight-of-evidence integration the pathway for using these methods without full formal validation. ARBITER produces exactly that integration, with a trace. We are not asking regulators to accept something new. We are building the thing their own guidance now depends on.

What a regulator would actually ask, and our answers

Is the reasoning reconstructable? Yes. Every verdict carries its rule chain, provenance, and version.

Where do the weighting rules come from? Published frameworks: Klimisch reliability scoring, OECD weight-of-evidence and IATA guidance, and AOP key-event confidence. Not our invention.

Who is accountable for the decision? The human expert, always. ARBITER produces the evidence package; the scientist signs it.

How do you handle a model outside its applicability domain? Rule R4 downweights or excludes it, and the system abstains rather than guessing.

13 Fitting Into the Existing Scientific Workflow

Where ARBITER physically sits, who touches it, and what changes. The answer to almost every adoption objection is that nothing upstream changes.

The workflow today

- 1 Assays are run and results land in separate systems: computational predictions in one place, in vitro results in a lab information system, omics in another, literature in people's heads and reference managers.
- 2 A safety lead manually pulls results together and assembles an evidence package for the review.
- 3 Where sources conflict, the lead reconciles them using experience and judgment. This reasoning is rarely written down in structured form.
- 4 The committee reviews, debates, and decides. The same conflicts are often re-derived from scratch in the meeting.
- 5 If the program advances, the reasoning is reconstructed later for the regulatory submission, often by different people.

The workflow with ARBITER

Steps 1, 4, and 5 are unchanged. ARBITER sits at steps 2 and 3, the assembly and reconciliation, which is exactly the part that is manual, inconsistent, and undocumented today. It reads from the systems already in place, produces a structured evidence package with the conflicts surfaced and reasoned through, and hands it to the same committee, who deliberate as they always have but from a clear starting point instead of a pile of disagreeing reports.

WHO USES IT DIRECTLY

The preclinical safety assessment lead. They open a compound, read the reasoning, adjust the rules where their expertise disagrees, and export the package.

WHO CONSUMES THE OUTPUT

The safety review committee, as a pre-read.
Regulatory affairs later, as the basis for the submission narrative.

WHO NEVER TOUCHES IT

Bench scientists generating the assay data. Their workflow is completely unchanged. This matters, because it means adoption does not require lab behavior change.

WHAT INTEGRATES

Reads from lab notebooks, information management systems, and assay databases. Wraps existing predictors rather than replacing them.

SAY THIS WHEN ASKED ABOUT ADOPTION

Nothing upstream changes. The same assays, the same systems, the same committee, the same decision-maker. What changes is that the reconciliation step, which today happens in someone's head and gets re-derived in every meeting, becomes structured, consistent, and written down.

PART 6 Patient Impact and Presentation

14 Pfizer and Patient Impact

A newly named judging criterion that our materials have never addressed. It is also, unexpectedly, one of our strongest sections, because our central case study is a story about patients.

The direct patient story

In the TAK-994 Phase 2 trial, eight participants exceeded predefined liver enzyme thresholds and three met Hy's Law criteria, the clinical marker for serious drug-induced liver injury. Those were real people with narcolepsy who volunteered for a trial. The evidence that would have flagged the risk existed in fragments across species and assays; it was never reconciled in time, and the mouse study that revealed the mechanism was only run once the trial was already underway.

That is the patient impact case, and it does not require exaggeration. **Better evidence reconciliation before first-in-human dosing means fewer trial participants exposed to avoidable harm.**

Four channels of patient benefit

Channel	How it works
Fewer patients exposed to unsafe candidates	Catching a reconcilable safety signal before first-in-human means the trial that would have injured participants never runs.
More safe medicines reaching patients	Overly conservative manual review kills borderline candidates that might have been viable. Naming the experiment that resolves the ambiguity rescues them, and a rescued medicine is a treatment someone eventually receives.
Resources redirected to candidates that can work	Every year and every dollar spent advancing a doomed compound is not spent on one that could help. Earlier termination is a patient benefit, not just a financial one.
Fewer animals used	Supporting the shift toward human-relevant methods advances the 3Rs directly, and human-relevant evidence is also better evidence for predicting human outcomes.

THE LINE FOR THE PATIENT IMPACT SLIDE

Three people in that trial developed serious liver injury. The signals that pointed to it existed, scattered across species and assays that disagreed with each other. Nobody had a structured way to put them together in time. That is the problem we are solving, and that is who we are solving it for.

15 The 15-Minute Presentation

Mapped to the guidelines' own recommended structure, business-weighted as our mentor instructed, with the demo at the center.

0:00 to 1:30

Problem

Open on TAK-994. A real drug, real patients, real harm, and every required study passed. Then the general problem: safety evidence conflicts routinely, reconciliation is manual and undocumented, and toxicity drives roughly a third of attrition. Name who experiences it: the preclinical safety lead.

1:30 to 3:00

Solution

The one-liner, then the five moves. Keep it conceptual. State plainly that the expert makes the decision. Land the differentiator: everyone predicts, nobody reconciles.

3:00 to 4:30

User and patient impact

The named persona with the before-and-after. Then the four patient channels, anchored back to the three Hy's Law cases.

4:30 to 6:00

Technical approach

High level only, per the guidelines and our mentor. The evidence streams, the reasoning layer, the uncertainty range, the data sources. Name key assumptions out loud. Do not go deeper than this; the demo carries the rest.

6:00 to 10:00

Demo

The centerpiece, four minutes. The nine screens from Part 2, ending on the consistency test. This is the section judges will remember and the one worth over-rehearsing.

10:00 to 11:00

Validation

The three baselines, the ablation, and the real numbers with sample size and intervals. State the pre-registration. This is where technical credibility is won.

11:00 to 13:00

Value to Pfizer

The business case: the three value channels, the asymmetry, the regulatory tailwind with the March 2026 guidance named, and workflow fit. Weight this heavily, per our mentor.

13:00 to 14:15

Feasibility and risks

Adoption path, cost, scalability, data security. Name the honest limitations before a judge does. This is a credibility move, not a concession.

14:15 to 15:00

Next steps and close

MVP status, near-term roadmap, long-term vision. Close on the mentor's one-liner, verbatim.

Presentation mechanics

- **All three of us speak.** The guidelines encourage it and the ambassador wants to know the team. Assign sections and rehearse the handoffs, which is where most teams look unprepared.
- **Rehearse to 14:00, not 15:00.** Live demos run long. Build in the buffer.
- **Rehearse the Q&A separately.** Five minutes with live judges is where the round is often decided. Use the question bank in Part 7.
- **Have the recorded demo walkthrough ready** as a fallback if anything fails live, and submit it with the materials.

PART 7 Risks, Q&A, and Schedule

16 Caveats, Assumptions, and Honest Limits

The guidelines explicitly ask whether assumptions are reasonable and well communicated. Stating these

ourselves converts our biggest vulnerability into a credibility signal.

Assumptions we are making

- That DILIrank classifications are an acceptable ground truth for hepatotoxicity. They are the FDA's own curated set, which makes this defensible, but they are classifications of known outcomes, not a perfect oracle.
- That public evidence streams are a reasonable proxy for the internal data a real deployment would use. Different in scale and quality, similar in kind.
- That our preference ordering, drawn from published frameworks, approximates what a practicing toxicologist would endorse. **This is our weakest assumption and we should say so.** The fix is expert review, which we are actively pursuing.
- That conflict frequency in our compound set is representative. With a small sample, it may not be.

Limits we will not paper over

- **ARBITER cannot invent signal that does not exist.** A first-in-class mechanism with no precedent in any source will not be predicted by any system, ours included. What ARBITER does is report wide uncertainty and route it to a human, which is the safe failure mode.
- **One case is an illustration, not proof.** The aggregate numbers are the claim.
- **Our sample is small.** Report intervals and say it plainly.
- **The demo does not prove enterprise security.** It runs on public data. The security architecture is a design commitment, not a demonstrated result.
- **Novelty is claimed as of mid-2026.** The agentic toxicology space is moving quickly. Re-check before presenting.

Verification checklist before August 16

Confirm every Pfizer-specific claim, including any executive quotes, against primary sources. Confirm the cost figures and present them as ranges with attribution. Re-run the competitive check on agentic toxicology tools. Remove anything that cannot be sourced. A Pfizer audience is exactly the audience that will catch an error about Pfizer.

17 Q&A Preparation

Five minutes with live judges. These are the questions most likely to come, with the short answer for each.

Are you replacing toxicologists?

No. The expert makes every decision. ARBITER assembles and reasons through the evidence so the human deliberates from a clear picture. The reasoning is symbolic specifically so a person can scrutinize and override every step.

Where do the weighting rules come from, and who says they are right?

Published frameworks: Klimisch reliability scoring, OECD weight-of-evidence and IATA, and AOP key-event confidence. They are explicit and editable, so a toxicologist can read, contest, and change them. We pre-registered them before evaluation so they were not fitted to the answer key.

How is this different from an LLM reading the same evidence?

We tested exactly that, and it is our ablation. The differences are consistency, since symbolic reasoning is deterministic and returns the same verdict every time, and traceability, since you read an argument rather than a score.

Pharma will not share data. How does this work in practice?

The MVP needs no proprietary data. In deployment, self-hosted open-weight models keep everything inside the environment, and because the reasoning core is symbolic rather than a trained network, it never needs to ingest anyone's compound library.

What if it kills a drug that would have worked?

ARBITER does not kill anything; people do, and the transparency makes a wrong call catchable rather than hidden. It also actively rescues borderline candidates by naming the experiment that resolves the doubt, which a conservative manual process would not do.

Pfizer has thousands of scientists. Why can they not already do this?

They do, by hand, and that is the problem. What is missing is consistency, scale, and a written record. We are encoding the experts' own standards and applying them the same way every time.

How long until this is real, and what does it cost?

Shadow mode within months, integrated over a longer horizon, with value from the first month because the evidence package is useful immediately. Cost is engineering and expert time, with no model training and no lab infrastructure.

How do you know it generalizes beyond liver?

We do not yet, and we scoped deliberately to one endpoint rather than claiming breadth we have not tested. The reasoning engine is endpoint-agnostic; extending means adding streams and pathway-specific rules. Cardiac is the natural next step.

What is your biggest weakness?

Answer honestly: the preference ordering has not yet been reviewed by a practicing toxicologist. It is drawn from published frameworks, but expert validation is the next thing we need, and we are pursuing it. A team that names its own gap is more trustworthy than one that claims none.

18 Schedule to August 16

Week of Jul 22	Data and rules. Assemble compounds, match identifiers by structure, normalize signals. Write and timestamp R1 through R6 before touching outcomes. Define conflict in code. Show the ambassador a working slice at the next check-in.
Week of Jul 29	Engine and baselines. Rules engine with trace output, fusion layer, and all three baselines including the language model pipeline. End-to-end results on the full compound set.
Week of Aug 5	Validation and interface. Run every metric, including repeat runs for consistency. Build the nine demo screens over precomputed results. Numbers locked by the end of this week.

Deck, rehearsal, submission. Build the business-weighted deck around real numbers. Record the fallback walkthrough. Rehearse to 14:00 with handoffs, then rehearse Q&A separately. **Submit by August 16, 11:59 PM ET, and submit early.**

The rule for every week

If something threatens the validation numbers or the demo working reliably, cut it. Those two are the submission. Everything else is supporting material.

19 Sources

Grounding claims in evidence is an explicit guideline instruction. Verify each before citing.

1. TAK-994 mechanistic investigation into drug-induced liver injury, *Toxicological Sciences* (2025), 204(2):143. The rat and primate studies missing the liability, the murine single-cell necrosis after CYP induction, and the wide in vitro margins.
2. Phase 2 TAK-994 results, *New England Journal of Medicine* (2023), and Takeda's program termination announcement. The 73 patients, the eight enzyme-threshold cases, and the three Hy's Law cases.
3. FDA, General Considerations for the Use of New Approach Methodologies in Drug Development, draft guidance, March 2026. The weight-of-evidence pathway.
4. FDA, Roadmap to Reducing Animal Testing in Preclinical Safety Studies, April 2025, and the FDA Modernization Act 2.0, 2022.
5. ICH M3(R2) and S9: the two-species repeat-dose requirement before first-in-human.
6. FDA DILIrank, Liver Toxicity Knowledge Base: ground-truth hepatotoxicity classifications.
7. Klimisch et al. (1997): study reliability scoring. OECD AOP handbook and AOP-Wiki: key-event confidence.
8. Tufts Center for the Study of Drug Development cost analyses; Martin et al. (2017) per-phase costs; Institute for Safe Medication Practices pivotal trial cost analysis. Present as a range.
9. DILI attrition and withdrawal literature, including reviews identifying hepatotoxicity as a leading cause of development termination and post-market withdrawal.

The closing thought

Round 1 shifts the question from "is this a good idea" to "can this realistically create value and be implemented." The guidelines say so directly. Everything in this document serves that shift. The demo proves it exists, the validation proves it works, the business case proves it is worth doing, and the regulatory landscape proves the timing is right. The idea is already good. The next four weeks are about making it real.